🔹 10 Easy-Level MCQs – IDS Architecture, Sensors, Agents, and DoS/DDoS

Q1. What is the purpose of a **network-based IDS sensor**?
A) Monitor disk usage
B) Capture and analyze network packets
C) Scan for viruses
D) Run OS updates
Answer: B
✅ *NIDS sensors are deployed to analyze real-time traffic for threats.*

Q2. A **DoS attack** aims to:
A) Steal user passwords
B) Block legitimate access to services
C) Encrypt files for ransom

D) Install antivirus updates
Answer: B

Q3. Which component in IDS **coordinates alerts and manages policies**?
A) Sensor
B) Agent
C) IDS Manager
D) Firewall
Answer: C

Q4. What is a symptom of a potential DDoS attack?
A) Slow network performance
B) Correct login attempts
C) Antivirus update
D) Scheduled backup
Answer: A

Q5. IDS agents are responsible for:
A) Changing IP addresses
B) Blocking email spam
C) Collecting and analyzing security data
D) Managing wireless routers

Answer: C

Q6. In a **tiered IDS architecture**, the lowest layer is:
A) User access control
B) Sensor/data collection
C) IDS manager
D) Alert engine
Answer: B

Q7. What type of attack involves **many systems** flooding a single target?
A) DoS
B) Brute-force
C) DDoS
D) XSS
Answer: C

Q8. One function of IDS sensors is to:
A) Format hard drives
B) Monitor and collect traffic
C) Host websites
D) Manage backups
Answer: B

Q9. The IDS component that **generates alerts** based on rules is:
A) IDS Manager
B) Firewall
C) Analyzer agent
D) Web proxy
Answer: C

Q10. What mitigation method can **rate-limit traffic** to prevent DoS?
A) DNS poisoning
B) IP spoofing
C) Rate limiting
D) VLAN hopping
Answer: C

Ready for the **20 Medium-Level MCQs**?

You said:

yes hard too

ChatGPT said:

Great! Let's continue with:

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20 Medium-Level MCQs – Sensors, Agents, DoS/DDoS, Architecture, IDS Manager

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Q11. Which component in a tiered IDS architecture is responsible for **correlation and policy enforcement**?
A) Sensor agent
B) IDS Manager
C) Host agent
D) Proxy server

Answer: B

Q12. Which of the following would MOST LIKELY indicate a sensor has detected an ongoing DoS attack?
A) High CPU usage on host
B) Decreased DNS resolution speed
C) Massive number of SYN packets to a port
D) Long-lived TCP connections
Answer: C

Q13. IDS **agents** typically do NOT:
A) Detect intrusions
B) Block traffic directly on the network
C) Analyze local logs
D) Communicate with central manager
Answer: B
Blocking is a function of IPS; IDS agents generally only alert.

Q14. What is the benefit of deploying **HIDS sensors** on endpoints?
A) Monitoring traffic at the perimeter
B) Centralized packet collection
C) Detecting unauthorized local changes
D) Handling firewall rules

Answer: C

Q15. What is one of the **main challenges** in deploying network-based sensors?
A) File system monitoring
B) Inability to detect encrypted payloads
C) Real-time log parsing
D) Host authentication
Answer: B

Q16. Why would an IDS deploy sensors in the **DMZ**?
A) To route traffic internally
B) To detect intrusions on exposed services
C) To monitor internet browsing
D) To decrypt emails
Answer: B

Q17. Which type of IDS sensor can detect a **rogue login shell** being spawned on a server?
A) Network sensor
B) Email filter
C) Host-based sensor
D) Load balancer
Answer: C

Q18. Which type of agent typically **initiates alerts to the manager**?
A) Firewall agent
B) Analysis agent
C) Logging agent
D) Alert agent
Answer: D

Q19. What does a **sensor agent** do?
A) Reboots hosts
B) Collects and forwards security data
C) Encrypts IDS logs
D) Assigns IP addresses
Answer: B

Q20. What is a **blackhole routing strategy** used for?
A) Monitoring network latency
B) Rerouting emails
C) Dropping malicious traffic silently
D) Bypassing firewall
Answer: C

Q21. Which of the following is LEAST likely a symptom of an attack?
A) New software installs without user action
B) Unusual outbound traffic
C) Decreased battery usage
D) User accounts being locked out
Answer: C

Q22. A tiered IDS architecture is MOST beneficial in:
A) Home networks
B) Isolated IoT devices
C) Large-scale enterprise deployments
D) Router firmware
Answer: C

Q23. A **scrubbing center** is used to:
A) Store IDS logs securely
B) Redirect and clean malicious traffic
C) Encrypt internet traffic
D) Train machine learning models
Answer: B

Q24. What is a **SPAN port** used for in sensor deployment?
A) Applying firewall rules
B) Sending copies of network traffic to IDS
C) Terminating VPN tunnels
D) Assigning public IPs
Answer: B

Q25. A sensor that uses **pattern recognition and traffic behavior** is applying:
A) Static rules
B) Anomaly detection
C) Port filtering
D) Session hijacking
Answer: B

Q26. Which function is MOST likely done by a **response agent** in IDS?
A) Save logs
B) Disable NIC
C) Forward alert to syslog
D) Kill a malicious process
Answer: D

Q27. A centralized **IDS Manager** typically runs:
A) On every host
B) Only during DoS attacks
C) On a central server collecting alerts
D) Inside every firewall
Answer: C

Q28. Which is a benefit of using multiple sensors in a distributed network?
A) Fewer logs to manage
B) Wider attack surface
C) Better coverage and redundancy
D) Reduced alert accuracy
Answer: C

Q29. An attacker is using **slow HTTP requests** to exhaust server threads. This is known as:
A) SYN flood
B) Slowloris attack
C) UDP flood
D) MAC spoofing
Answer: B

Q30. Which type of IDS component would be deployed on a database server to track unauthorized queries?

A) Network firewall
B) Host-based IDS sensor
C) DNS server
D) Proxy agent
Answer: B

🔴 10 Hard-Level MCQs – Deployment, Architecture, Threat Detection

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Q31. What could happen if sensors are deployed only at the network perimeter?
A) All internal attacks are detected
B) Only encrypted traffic is analyzed
C) Insider threats may go undetected
D) Hosts become unbootable
Answer: C

Q32. Which detection technique is MOST effective for identifying **zero-day denial-of-service** attacks?
A) Signature-based
B) Static packet filtering
C) Anomaly-based detection
D) Protocol enforcement
Answer: C

Q33. An agent that uses **Syscall hooking** is most likely:
A) Network-based
B) Wireless sensor
C) Host-based
D) Protocol translator
Answer: C

Q34. The ideal placement for detecting **lateral movement** in a network is:
A) Firewall
B) DMZ sensor
C) Core/internal switch sensor
D) DNS server
Answer: C

Q35. Which component is responsible for **aggregating alerts** and generating a **unified threat timeline**?
A) Sensor agent
B) IDS Manager
C) Alert buffer
D) Response proxy
Answer: B

Q36. Why is **sensor overloading** a risk in large IDS deployments?
A) Leads to IP conflicts
B) Creates false negatives from dropped packets
C) Stops antivirus updates
D) Overwrites router configs
Answer: B

Q37. A **multi-tier IDS** with distributed sensors and centralized management is an example of:
A) Flat topology
B) Hub-and-spoke
C) Hierarchical architecture
D) Asynchronous design
Answer: C

Q38. Which method is BEST for detecting attacks **that evolve over time**?
A) Snapshot analysis
B) Deep packet inspection
C) Event correlation across sensors
D) Random scanning
Answer: C

Q39. A sensor reports thousands of outbound TCP connections from a workstation. This may indicate:
A) Hardware upgrade
B) Port forwarding
C) Botnet behavior
D) High system uptime
Answer: C

Q40. To ensure agents are not easily disabled by attackers, a best practice is to:
A) Run them as user-level services
B) Install them on routers
C) Encrypt agent communications and run as system-level daemons
D) Disable real-time monitoring
Answer: C